Assessment of eddy resolving techniques for the flow over periodically arranged hills up to Re=37,000

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Summary. The turbulent flow over periodically arranged geometrically two-dimensional hills in a channel at a Reynolds number of $Re = 37,000$ has been considered as benchmark case for various eddy-resolving methods. The aim of this study is to assess various LES models and numerical approaches in a turbulent flow detaching from a curved surface. We compare results of a Cartesian grid solver using the immersed boundary method with various curvilinear approaches ranging from standard eddy-viscosity subgrid-scale models to hybrid LES-RANS models. The results are validated by a recent experiment conducted in a water channel by particle image velocimetry and laser-Doppler anemometry.

Key words: Quality, Reliability, Large-Eddy Simulation, Hybrid LES-RANS

1 Introduction

Large-Eddy Simulation of complex aerodynamic configurations heavily stresses available computer hardware and codes through excessive grid refinement necessary to resolve the immediate wall layer. The scaling of the viscous layer with Reynolds number leads to a severe bottleneck if flows at high Reynolds numbers are to be predicted. Furthermore, complex high Reynolds number flows are difficult to predict because of the representation of the geometric details. Concerning the representation of the geometry, several grid systems have been developed, Cartesian Immersed Boundary (IBM), Curvilinear Body Fitted and unstructured grids. In this study, we consider the former two by comparing results from three different flow solvers, a Cartesian IBM code (MGLET, [11]) a curvilinear code that allows for a hybrid method (LESOCC, [2]) and a curvilinear code with a recently developed wavelet-based subgrid-scale model (LESOCC2, [7]).

While IBM methods are able to represent highly complex shapes with many geometric details, they have problems to refine the grid close to a wall if the wall is not

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aligned with the grid. This problem results in poor wall shear stress predictions if not two (or even three in the general case) directions are refined at the wall. Consequently, it has been found in a previous study at a low Reynolds number that an IBM method requires a larger total number of grid cells than a curvilinear one. We expect that with increasingly thinner viscous sublayers, i.e. with higher Reynolds numbers, this problem would be more severe. The performance of an IBM method together with three different sub-grid scale models is checked by the flow solver MGLET. We compare standard Smagorinsky [13], the WALE subrid-scale model [10] and a Lagrangian dynamic model [9].

As an alternative to explicit wall functions, hybrid methods have become popular in recent years. Hybrid methods, such as Detached Eddy Simulation (DES) promise to save resources in the wall layer by applying Reynolds Averaged (RANS) models in the wall layer and LES outside. Since at high Reynolds numbers, wall units can not be resolved by a grid that resolves the large scales of the outer flow, the spatial filter performed by this grid tends towards a Reynolds filter thus justifying RANS modeling close to the wall. Issues of these so-called hybrid methods are the location and definition of the interface between RANS and LES zone and the low-Reynolds modification of the terms in the applied model equations. A recent development of Jaffrezic and Breuer [6] showed excellent results for the considered flow case at a lower Reynolds number. We assess this approach, a combination of Schumann's one-equation subgrid-scale model in the LES zone with an explicit algebraic Reynolds stress model in the RANS zone in comparison with a pure LES with Schumann's model by the curvilinear code LESOCC.

A recently proposed wavelet-based eddy-viscosity subgrid-scale model [4] is checked in comparison to the standard Smagorinsky model with the curvilinear code LESOCC2. In this model the eddy-viscosity is set proportional to the wavelet-details from a two-level biorthonormal wavelet decomposition of the velocity vector.

2 Test case

We consider in this paper the flow over a periodic arrangement of smoothly contoured two-dimensional hills featuring important details of flow separation from a curved surface with subsequent reattachment (see Figure 1). It goes back to the experiments of Almeida *et al.* [1], but has been modified by Mellen *et al.* [8] to be more suitable for numerical simulations. Consequently, it has been extensively studied over the past few years e.g. [3, 5] and used as basis for workshops on turbulence modeling. Up to now, highly resolved LES and DNS are available for Reynolds numbers $Re \leq 10,595$ that can be regarded as reference solutions [3]. In addition, an experiment corresponding to this new setup was designed at the Technische Universität München in order to provide experimental reference data up to $Re = 37,000$ [12]. In this experiment, careful checks have been made to validated the measurement techniques (PIV and LDA). Two-dimensionality of the flow in the middle of the channel has been proven for $Re \geq 10,595$.

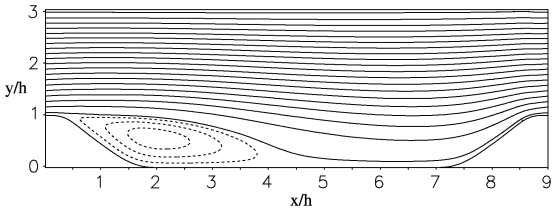

Fig. 1. Geometry of periodic hills in a channel.

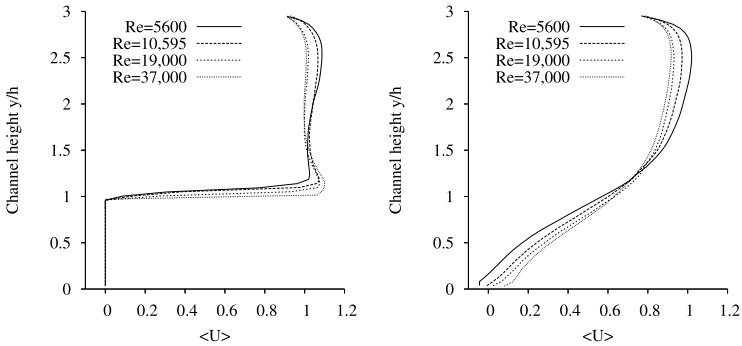

Fig. 2. Dependence of averaged velocity profiles on Reynolds number. Experiment by Rapp [12]. $x/h = 0.05$ (left) and $x/h = 4.0$ (right).

The flow field reveals all structural features of separating and reattaching flows including curved streamlines, highly unsteady separation and reattachment lines, a fluctuating shear layer above the separation zone giving rise to Kelvin-Helmholtz vortices and strongly amplified fluctuations. On the windward side of the hill the flow is accelerated with a maximal wall shear stress occurring approximately in the upper third of the hill slope. Two secondary separation zones were observed at $Re = 10,595$, one in front of the hill and one at the hill crest. A detailed analysis of the flow field was reported by Fröhlich *et al.* [5] and Breuer *et al.* [3].

Of special interest for this study is the Reynolds number dependence of the flow field. As can be seen in Figure 2, the mean streamwise velocity profiles develop an overshoot directly above the hill crest ($x/h = 0.05$) with increasing Reynolds number. Above $Re = 19,000$ the profile is flat towards the upper wall over the hill. The separation point moves upwards with increasing Reynolds number. Close to the reattachment point ($x/h = 4.0$) we see again the flatter velocity profiles at the higher Reynolds numbers. The two positions shown in Figure 2 reveal the most pronounced Reynolds number dependent features of the averaged velocity field. These are also the positions at which the strongest deviations between different simulations have been observed in previous studies.

3 Numerical details

The codes used in this assessment have been developed for LES of turbulent flows and validated in a number of studies. All codes use finite-volume discretizations with standard central second-order spatial approximations for this study and Runge-Kutta time advancement. The purpose of this study is to assess what can be achieved with relatively coarse grids. In this context, it is of major interest, how the performance of the schemes changes with increasing Reynolds number.

The two curvilinear codes use grids that are refined towards the wall and result in 10^6 (LESOCC2) and $1.5 \cdot 10^6$ (LESOCC) cells, respectively. The most critical point concerning resolution is at the maximal wall shear stress at the windward side of the hill. The spacings Δy^+_{max} that could be achieved by these grids are rather coarse for $Re = 37,000$ (see table 1). For the Cartesian method it is even more difficult to achieve a full wall resolution at that point with a comparable number of cells. We performed a resolution study with two different grids, one with $2 \cdot 10^6$ (2M), the other with $3.8 \cdot 10^6$ (4M) cells. The achievable wall resolutions at the point of maximal wall stress are very coarse. However, it has to be noted that the length over which this wall stress is acting upon the flow field is very short, approximately one half hill heights. In all other positions, the wall resolutions can be regarded as sufficient.

Table 1. Main parameters of grids used in this study. X: streamwise; Y: wall normal; Z: spanwise direction. Wall resolution Δy^+_{max} at point of maximal wall shear stress.

Code	$NX \times NY \times NZ$	$\Delta y^+_{max}(10,595)$	$\Delta y^+_{max}(37,000)$
MGLET (4M)	$216 \times 168 \times 104$	11	30
MGLET (2M)	$176 \times 120 \times 96$	32	88
LESOCC	$160 \times 160 \times 60$	1	3.4
LESOCC2	$160 \times 100 \times 60$	4.3	11.8

4 Results

The focus of this study lies on the capability of the different modeling approaches to accurately predict the size and shape of the separation downstream of the hill at a medium Reynolds number. It can be stated that all methods succeed in qualitatively predicting the separation zone. The main difference between the various models and approaches can be seen in the time averaged streamwise velocity profiles at the hill crest $x/h = 0.05$ and the reattachment region $x/h = 4.0$ which are discussed in the remainder.

The performance of the IBM method is discussed for $Re = 10,595$ and $Re = 37,000$ (Figures 3 and 4). The overall accordance is satisfying. The peak over the hill crest that was measured in the experiment is not reached by the simulations

at $Re = 10,595$. Especially the Smagorinsky model (SMA) seems to underpredict this feature. This underprediction goes in hand with a delayed reattachment (see $x/h = 4.0$). The Lagrangian subgrid-scale model (LAG) performs best at the lower Reynolds number. At the higher Reynolds number, the WALE model (WAL) performs a little better than the Lagrangian one but both are fully satisfying over the hill crest and at the point of reattachment. The Lagrangian model with $2 \cdot 10^6$ (2M) cells even outperforms the Smagorinsky model at twice the number of cells. But both reattach too late. The comparison between experiment and simulation is better at the higher Reynolds number than at the lower one. This is surprising as the resolution of the viscous layer is much worse at the higher Reynolds number and no wall model was used.

The performance of the curvilinear codes is checked at $Re = 37,000$ (Figure 5). Concerning the overshoot over the hill crest there are two different solutions, the finer grid (1.5M) comes closer to the experiment than the coarser one (1M). It seems that a too small overshoot over the hill crest goes in hand with a too long reattachment length. The hybrid model (HYB) performs similar to Schumann's model (SCH) with a slightly longer recirculation zone. The wavelet-based model (WAV) is slightly better than the Smagorinsky model. To check the performance of the curvilinear schemes at $Re = 10,595$, we plot profiles from Smagorinsky (LESOCC2, 1M-SMA), wavelet-based (LESOCC2, 1M-WAV) and dynamic Smagorinsky (LESOCC, 12M-DSM). Note, that the latter one was computed on a $12 \cdot 10^6$-cells mesh that is fine enough so that the effect of SGS models disappears [3]. This simulation can be regarded as reference for LES at this Reynolds number as no significant change of the results can be expected at higher resolutions. At this Reynolds number, the standard Smagorinsky model performs nearly as well on one million as the dynamic one on 12 million cells. The wavelet-based model is a little too dissipative at the wall as no damping of the SGS-viscosity was undertaken. This issue has already been identified for further improvements of this model. Note that Van Driest damping was used for the Smagorinsky model.

The assessment of the averaged velocity profiles over the hill crest and close to reattachment reveals the observation that an underprediction of the overshoot above the hill crest results in a longer recirculation zone in the lee of the hill. This behavior can not fully be explained by the values of the total shear stresses (resolved plus modeled) at the hill crest (Figure 7). The higher Reynolds shear stress values come together with a longer separation bubble. This is counterintuitive but can be explained by the fact that the momentum transport over the shear layer bounding the separation zone is due to the fluctuations in that shear layer which start over the hill crest in a very thin layer. This layer is represented by the sharp shear stress peak immediately above the wall that was not measured in the experiment, presumably due to resolution problems. Thus the large peak in the middle of the channel is a remainder of upstream processes. It can be concluded that it is responsible for the shape of the velocity profile in the bulk of the flow which was observed to be more or less flat in the various simulations. If the shear stresses are strong, the overshoot just over the crest of the hill is damped due to increased momentum transfer from the bulk to the location of the overshoot. This conclusion is supported by the fact that all results with lower peak

(overshoot) values directly over the hill crest show an increased level of Reynolds shear stress in the bulk.

Now let's examine the behavior of the individual methods upstream of the hill which is supposed to be important for the shape of the velocity profile above the crest. The windward side of the hill is characterized by a strong acceleration of the bulk velocity by a factor of 3/2. An examination of the Reynolds shear stresses at the middle of the hill slope at $x/h = 8.0$ reveals a nearly perfect match of all methods with the measurements at $y/h \geq 1.5$ (not shown). In this region the streamlines are nearly parallel (compare Figure 1). Below, the numerical results deviate from the experiment. Reaching towards the lower wall, the streamlines are more and more aligned with the hill slope. The momentum transfer perpendicular to the streamlines can therefore only be examined in a coordinate system that is aligned with the wall which is done in the following.

The Reynolds number dependence of the shear stress that is evident from the experiment is hardly predicted by the numerical schemes (Figures 8 and 9). Note that in these figures, on the left hand side the numerical results should coincide with the lower Reynolds number (hollow symbols) and on the right hand side, they should match the higher Reynolds number (full symbols). While in the experiment the shear stress drops by a factor of $3/2$ from $Re = 10,595$ to $Re = 37,000$, this drop is not observed in most of the models. In the Cartesian code, the Lagrangian and the WALE models drop by some amount but not like the experimental values. At the lower Reynolds number, the values are mainly too small, at the higher one too large. The results from the Smagorinsky model seem to be independent of the Reynolds number, an observation that can also be made in the curvilinear codes. Those results lie generally above the experimental ones except for the high resolution simulation with $12 \cdot 10^6$ cells (12M-DSM) at $Re = 10,595$. The main trend is that the higher the momentum transport perpendicular to the streamlines is, the lower is the overshoot just above the hill crest. This is explained by the fact that this overshoot is an inviscid effect that is attenuated by strong shear stresses upstream. The excessive near-wall bump apparent in the IBM method is a consequence of the poor near-wall resolution of this method. Its effect on the velocity profiles remains limited as the overall level remains small.

5 Conclusions

The present study compares several eddy resolving approaches for numerical simulation of turbulent flow with experimental data at low to medium Reynolds number flow over periodically arranged hills. Special attention is placed on the Reynolds number dependence of the results. The following observations could be done.

The acceleration and streamline curvature at the windward side of the hill generates an overshoot in the velocity profile above the hill crest. This is an inviscid effect. Viscous and turbulent stresses attenuate this overshoot. Both, viscous and turbulent stresses decrease with increasing Reynolds number, consequently the overshoot increases with Reynolds number. Various methods were not able to reproduce

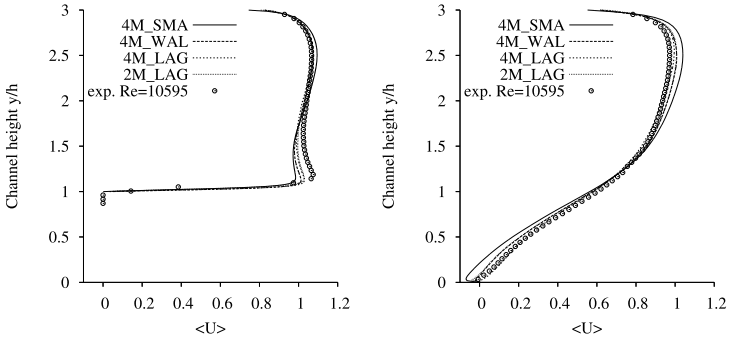

Fig. 3. Comparison of averaged velocities from MGLET with experimental data at $Re = 10,595$. Experiment by Rapp [12]. $x/h = 0.05$ (left) and $x/h = 4.0$ (right).

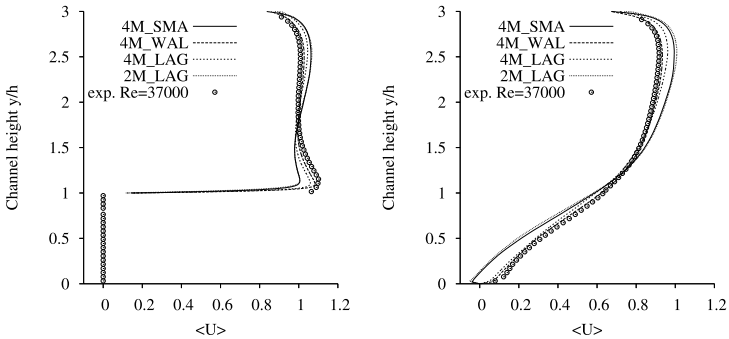

Fig. 4. Comparison of averaged velocities from MGLET with experimental data at $Re = 37,000$. Experiment by Rapp [12]. $x/h = 0.05$ (left) and $x/h = 4.0$ (right).

this Reynolds number dependence, especially the classical Smagorinsky model. The successful models include the Lagrangian dynamic, the WALE and the newly developed wavelet-based model as well as Schumann's model with or without hybrid wall modeling. It has to be mentioned that the hybrid wall modeling tested in this study represents a promising way as the interface between RANS and LES zone does not have to be defined a priori and a smooth transition is achieved at the interface. Van Driest damping is required when the wall resolution is in the range of ten wall units. The newly proposed wavelet-based model still has to be tested with Van Driest damping although it's results are fully satisfying.

A correct prediction of the maximum wall shear stress, which is difficult for IBM methods seems to have less impact on the global results at higher Reynolds numbers than at lower ones. This can be explained by the ratio of wall shear stress to wall pressure which shifts in favor of the wall pressure at higher Reynolds numbers. As a consequence, results at higher Reynolds number are less prone to immediate wall resolution than to overall grid resolution in the bulk of the flow. In this respect, the penalty of Cartesian IBM methods concerning wall resolution is less severe for this flow when Reynolds number is high.

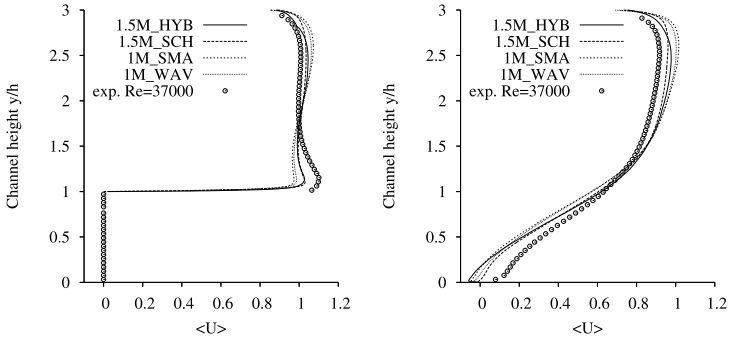

Fig. 5. Comparison of averaged velocities from LESOCC and LESOCC2 with experimental data at $Re = 37,000$. Experiment by Rapp [12]. $x/h = 0.05$ (left) and $x/h = 4.0$ (right).

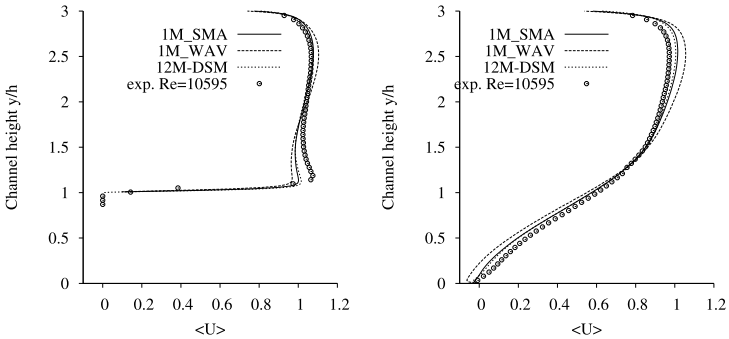

Fig. 6. Comparison of averaged velocities from LESOCC and LESOCC2 with experimental data at $Re = 10,595$. Experiment by Rapp [12]. $x/h = 0.05$ (left) and $x/h = 4.0$ (right).

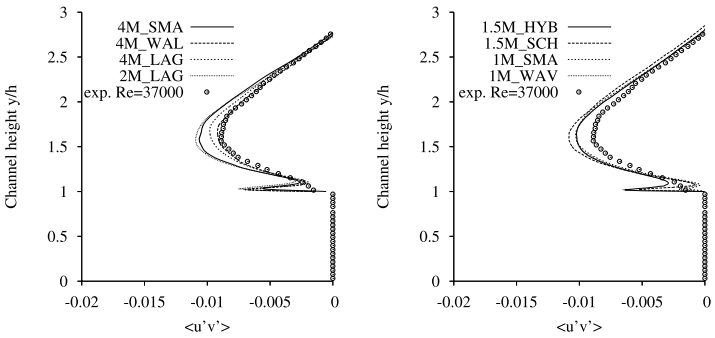

Fig. 7. Comparison of Reynolds shear stress from MGLET (left) and LESOCC and LESOCC2 (right) with experimental data at $Re = 37,000$. Experiment by Rapp [12]. $x/h = 0.05$.

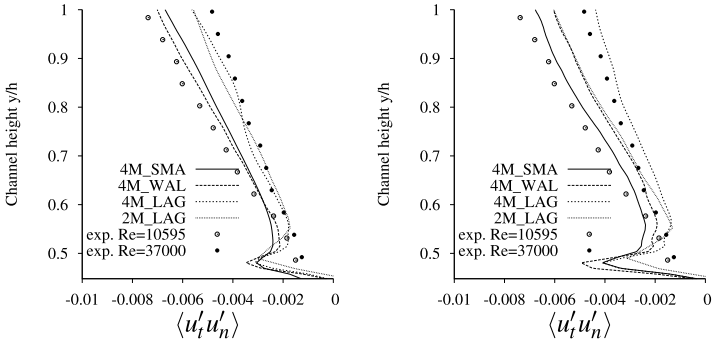

Fig. 8. Comparison of Reynolds shear stresses from MGLET at $x/h = 8.0$ in a coordinate system aligned with the wall. Experiment by Rapp [12]. $Re = 10,595$ (left) and $Re = 37,000$ (right).

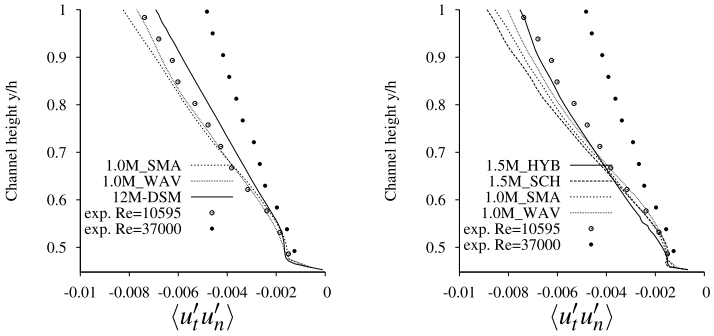

Fig. 9. Comparison of Reynolds shear stresses from LESOCC and LESOCC2 at $x/h = 8.0$ in a coordinate system aligned with the wall. Experiment by Rapp [12]. $Re = 10,595$ (left) and $Re = 37,000$ (right).

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